

# Tianyi Zhang

+1 404-422-2842 | tzhang708@gatech.edu | LinkedIn | GitHub | Google Scholar | tianyi-zhang-02.github.io

## EDUCATION

### Georgia Institute of Technology

M.S. in Computer Science

Atlanta, GA

Aug 2025 – May 2027

### Emory University

B.S. in Data Science

Atlanta, GA

Aug 2021 – May 2025

## PROFESSIONAL EXPERIENCE

### LinkedIn Corporation

Machine Learning Engineer Intern

Mountain View, CA

May 2026 – Aug 2026

- Improved the training and evaluation foundations of a production **two-tower retriever**: fixed data and loss implementation issues and identified supervision dominated by reading time, missing rejection feedback, and evaluation negatives inferred from unexposed content.
- Designed and handed off a **Qwen multi-tower architecture** that learns user profile and interaction history separately, with learnable routing across their candidate pools; defined tests for unique relevant candidates under a fixed retrieval budget.
- Designed an offline **VLM-teacher / embedding-student architecture** to enrich text with visual evidence and generate relevance supervision and hard negatives; proposed precomputed candidate embeddings so multimodal understanding adds no per-request VLM inference.
- Shipped a **PySpark evaluation framework over 2.7B member-candidate pairs**, comparing relevance, coverage, and topical diversity across matched user cohorts; found Semi-cold users had about **25% higher Recall@10 but 22% lower ILD@10** than Active users, revealing accurate but less diverse retrieval.

### Georgia Tech EIC Lab · NVIDIA AI Research

Student Researcher

Atlanta, GA

Jul 2025 – Dec 2025

- Engineered the **AgentSuite** benchmark-auditing pipeline with a taxonomy-guided LLM judge and deterministic structural detectors, matching expert audits at **0.79–0.87 F1** across **6 benchmarks**.
- Generalized to unseen benchmarks without retuning; identified flaws in **23% of tasks** whose correction changed **63% of model rankings**.

### Emory Graph Mining Lab

Machine Learning Researcher

Atlanta, GA

Dec 2023 – May 2025

- Designed a **Granger-causal DAG framework** to model directional dependencies in fMRI time series rather than undirected correlations, using the resulting brain networks to improve downstream patient diagnosis.
- Co-developed a **Brain Network State Transformer** that groups changing brain connectivity into recurring states and learns patterns within and between states, capturing temporal information lost in static network models.

## OPEN SOURCE

### NVIDIA NeMo RL (2.0K stars)

Merged PRs  Open PRs 

Open-Source Contributor | Scalable toolkit for efficient LLM reinforcement learning and post-training.

- Performance & Correctness**: Optimized log-probability and top- $k$  distillation paths, making cross-token P-KL **3.7× faster** and reducing tensor-parallel communication from **250 MiB to 16 MiB per rank**.
- Multi-Reward GDPO**: Developing end-to-end preservation of named reward components across async rollout, replay, checkpoint/restore, and custom policy objectives for composable multi-reward post-training.
- SingleController PPO**: Developing streamed actor and critic minibatches to avoid materializing merged payloads, targeting lower memory pressure while preserving functional resume and training parity on H100s.

## PUBLICATIONS

H. Suh\*, S. Lee\*, B. Ji\*, R. Khare, B. Khan, H. Kim, **T. Zhang**, et al. “AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline” *International Conference on Machine Learning (ICML)* 2026.

**T. Zhang**, K. Han, J. Nie, et al. “Causal Brain Connectivity: Integrating Granger Directed Graphs in fMRI Analysis.” *International Conference on Artificial Intelligence in Medicine (AIME)* 2025.

J. Nie, K. Han, **T. Zhang**, et al. “Brain Network State Transformer: Leveraging State Functional Connectivity.” *IEEE Engineering in Medicine and Biology Society (EMBC)* 2025. (Oral)

## TECHNICAL SKILLS

|               |                                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Languages     | Python, Go, SQL, C/C++, Java, R                                                   |
| ML Frameworks | PyTorch, TensorFlow, PyTorch Lightning, Hugging Face Transformers                 |
| Post-Training | NVIDIA NeMo RL, verl, vLLM, SGLang, CUDA                                          |
| Data & Tools  | PySpark, Apache Spark, Flyte, MLflow, FastAPI, Docker, AWS, GitHub Actions, Linux |